DAY 1 – NETWORK INVESTIGATION & INFRASTRUCTURE

Focus: OSI/TCP-IP reasoning, IP configuration, routing, DHCP, NAT, DNS, and diagnostic tools.

PART A – Network Architecture & Reasoning

1. A developer opens a website from a laptop connected to office Wi-Fi. Describe the journey from the laptop to the website server and back. Include the roles of the network interface, default gateway, ISP, DNS, routing, and the web server.

Journey of a Laptop Request to a Website

- **When a developer opens a website from a laptop connected to office Wi-Fi, the following steps occur:**
- **Network Interface: The laptop uses its Wi-Fi network interface to connect to the local network.**
- **DNS: DNS translates the website's domain name into an IP address.**
- **Default Gateway: The laptop forwards the packet to the default gateway (router) because the destination is outside the local network.**
- **ISP: The Internet Service Provider forwards the traffic towards the internet.**
- **Routing: Routers forward packets through different networks until they reach the destination server.**
- **Web Server: The web server receives the HTTP/HTTPS request and sends the requested website content.**
- **Response: The response travels back through the network to the laptop, and the browser displays the webpage.**

2. A team says, “The OSI model and TCP/IP model are the same.” Explain why this statement is incomplete. Map the
seven OSI layers to the TCP/IP model and give one practical protocol or example for each OSI layer.

Difference Between OSI and TCP/IP Models

The OSI and TCP/IP models are not exactly the same. The OSI model is a conceptual model with seven layers, while the TCP/IP model is a practical networking model generally organized into four layers.

Mapping:

- Application, Presentation, Session → Application Layer (HTTP, FTP, TLS, RPC)
- Transport → Transport Layer (TCP, UDP)
- Network → Internet Layer (IP, ICMP)
- Data Link, Physical → Network Access Layer (Ethernet, Wi-Fi, physical cables and signals)
- The OSI model helps understand network communication in detail, while the TCP/IP model is widely used in real-world internet communication.

3. A request reaches the server, but the browser never displays the page. Explain how a problem at the Physical,
Network, Transport, or Application layer could produce a similar user symptom. Give one possible fault at each layer.

Network Layer Problems That Prevent a Webpage from Displaying

A website may fail to display due to problems at different networking layers.

- **Physical Layer:** A damaged network cable or Wi-Fi connection failure may interrupt network communication.
- **Network Layer:** An incorrect IP address or routing problem may prevent packets from reaching the destination server.

- **Transport Layer:** A firewall blocking TCP port 443 may prevent an HTTPS connection from being established.
- **Application Layer:** The web server may return an internal server error (HTTP 500), preventing the browser from displaying the expected page.

Although the faults occur at different layers, the user may experience the same symptom: the webpage does not load.

4. An application runs on a developer laptop, a Docker container, and a cloud VM. Explain why these environments may use different IP addresses and how communication between them is established.

Different IP Addresses in Laptop, Docker and Cloud VM

A developer laptop, Docker container and cloud VM may have different IP addresses because they operate in separate network environments.

- **Developer Laptop:** Usually receives a private IP address from the local Wi-Fi router through DHCP.
- **Docker Container:** Docker assigns a separate private IP address using its virtual network.

- **Cloud VM:** The cloud provider assigns a private IP address within its virtual network. It may also have a public IP address for external access.
- Communication between these environments is established using routing, NAT, Docker port mapping, security groups and firewall rules.

Therefore, different environments can have different IP addresses while still communicating with each other through appropriate network configuration.

PART B – Linux Network Forensics

Run the following commands and capture their outputs: ip -br address, ip route, ip neigh, cat /etc/resolv.conf, hostname -I, and curl -s https://api.ipify.org.

5. From the output, identify the active network interface, assigned IPv4 address, subnet/CIDR (if shown), default route, DNS resolver, and public-facing IP. Explain which values belong to the local network and which are externally visible.

Linux Network Configuration Investigation

The following commands were used to investigate the Linux network configuration:

- ip -br address – Identifies network interfaces, their status and assigned IP addresses.
- ip route – Displays the routing table and default gateway.
- ip neigh – Displays neighboring devices and their IP and MAC addresses.
- cat /etc/resolv.conf – Identifies the configured DNS resolver.
- hostname -I – Displays the IP addresses assigned to the system.
- curl -s https://api.ipify.org – Identifies the public-facing IP address.

Network Information:

- Active network interface: To be identified from actual output.
- IPv4 address: To be identified from actual output.
- Subnet/CIDR: To be identified from actual output.
- Default gateway: To be identified from actual output.
- DNS resolver: To be identified from actual output.
- Public IP: To be identified from actual output.

Local IP addresses are used for communication within private networks, whereas the public IP address is visible to external internet services.

6. Use ip route get 1.1.1.1. Explain how the output helps identify the interface and next hop selected for an
outbound packet.

6. Route Selection for an External IP

Command used:

ip route get 1.1.1.1

Explanation:

The ip route get command shows the route selected by the Linux kernel to reach a specific destination IP address.

It helps identify the following details:

- **Destination IP: The IP address to which the packet is sent.**
- **Next Hop: The gateway through which the packet is forwarded.**
- **Network Interface: The outgoing interface used to send the packet.**
- **Source IP: The source IP address selected for the outgoing packet.**

For example, if the output contains via 172.25.64.1 dev eth0 src 172.25.64.10, it means the packet will be forwarded through gateway 172.25.64.1, using interface eth0, with source IP 172.25.64.10.

Conclusion:

This command helps understand how Linux selects the outgoing network interface and next-hop gateway when sending packets to an external network.

7. Use ping -c 4 1.1.1.1 and ping -c 4 google.com. Compare the results. What can you infer if the first
succeeds but the second fails?

Ping Connectivity Test

Commands used:

ping -c 4 1.1.1.1
ping -c 4 google.com

Explanation:

The first command checks network connectivity by sending four ICMP packets directly to the public IP address 1.1.1.1. DNS resolution is not required.

The second command sends four ICMP packets to google.com. Before sending the packets, the system uses DNS to resolve the domain name into an IP address.

Comparison:

ping 1.1.1.1 checks connectivity to a destination using its IP address.

ping google.com checks DNS resolution as well as network connectivity.

Inference:

If ping to 1.1.1.1 succeeds but ping to google.com fails, it may indicate a DNS resolution problem. However, ICMP filtering or firewall restrictions can also cause ping failures.

Conclusion:

Ping is a useful troubleshooting tool for checking network reachability and identifying possible DNS-related issues.

8. Use traceroute google.com or tracepath google.com if available. Explain what each hop represents and
why some hops may show asterisks or no response.

Traceroute Network Path Investigation

Command used: traceroute google.com

Explanation:

Traceroute is a network diagnostic tool used to identify the path taken by packets from the source system to a destination server.

Each hop represents an intermediate router or network device that forwards packets towards the destination.

The output displays the hop number, responding router IP address and response time.

Why do some hops show asterisks (*)?

Asterisks indicate that a hop did not respond within the specified timeout period. Possible reasons include firewall restrictions, packet filtering, network congestion or routers configured not to respond to traceroute probes.

Asterisks do not necessarily indicate a network failure because the router may still forward packets without responding to traceroute requests.

Conclusion:

Traceroute helps identify the network path and investigate possible routing delays or connectivity problems between the source and destination.

9. Use ss -tuln. Identify at least three listening ports, if present. Explain the difference between a listening socket and
an established connection.

Identify Listening Ports in Linux

Command used: ss -tuln

Explanation:

The ss -tuln command displays TCP and UDP listening sockets on a Linux system. The -t option displays TCP sockets, -u displays UDP sockets, -l displays listening sockets, and -n displays numerical addresses and port numbers.

The output helps identify which services are listening on the system and the local addresses and ports on which they are available.

Observation:

I executed the command on my Ubuntu system and observed the listening sockets and their associated protocols, states, IP addresses, and port numbers.

PART C – IP Addressing, DHCP, Gateway & NAT Scenarios

10. A workstation has IP 192.168.20.45/24 and gateway 192.168.20.1. State the network address, usable host
range, and broadcast address. Show how you derived them.

Network Address, Usable Host Range and Broadcast Address

Given:

IP Address: 192.168.20.45/24

Gateway: 192.168.20.1

A `/24` prefix corresponds to the subnet mask `255.255.255.0`.

Calculation:

Network Address: 192.168.20.0

First Usable Host: 192.168.20.1

Last Usable Host: 192.168.20.254

Broadcast Address: 192.168.20.255

The first 24 bits represent the network portion, while the remaining 8 bits represent the host portion. Setting all host bits to 0 gives the network address, and setting all host bits to 1 gives the broadcast address.

Final Answer: The usable host range is 192.168.20.1 to 192.168.20.254. The given gateway, 192.168.20.1, is within this range.

11. Two servers are configured as 10.10.5.12/24 and 10.10.6.18/24. Are they in the same IPv4 subnet? Explain
whether they can communicate directly or need routing.

Given:

Server 1: 10.10.5.12/24

Server 2: 10.10.6.18/24

With a `/24` subnet mask, the network addresses are:

Server 1 Network: 10.10.5.0

Server 2 Network: 10.10.6.0

The two network addresses are different. Therefore, the servers are in different IPv4 subnets.

They cannot communicate directly at Layer 2. Communication requires a router or Layer 3 gateway with appropriate routing and firewall rules.

12. A laptop receives an address beginning with 169.254. What does this commonly indicate? Give a troubleshooting
sequence to investigate the DHCP failure.

Troubleshooting a 169.254.x.x IP Address

An IP address beginning with 169.254 commonly indicates that the device could not obtain an IP address from a DHCP server. The operating system may assign itself an APIPA (Automatic Private IP Addressing) address.

Troubleshooting sequence:

- Check the Ethernet cable or Wi-Fi connection.
- Verify the IP address and routing configuration.
- Check whether the DHCP server is reachable and operational.
- Renew the DHCP lease or reconnect the network interface.
- Check DHCP logs, available address scope, VLAN configuration, and DHCP relay settings.
- After resolving the DHCP issue, verify that the device receives a valid IP address, subnet mask, gateway, and DNS configuration.

13. A private server can access the internet, but an internet user cannot initiate a connection to that server. Explain how
NAT and firewall rules may account for this behavior.

A private server can access the internet because the router or firewall commonly uses Source NAT (SNAT) or Port Address Translation (PAT) to translate its private source IP address into a public IP address.

However, an internet user cannot directly initiate a connection to the private server because private IP addresses are not directly routable over the public internet.

To allow inbound connections, the network administrator may need to configure Destination NAT (DNAT) or port forwarding and appropriate firewall rules. The server must also be running the required service and listening on the specified port.

14. A cloud VM has a private IP and a public IP. Explain the purpose of each and why the private IP is commonly used for communication inside a cloud network.

A cloud virtual machine may have both a private IP address and a public IP address.

Private IP: Used for communication between resources inside the same cloud network or connected private networks. It is not directly reachable from the public internet.

Public IP: Used to enable communication between the VM and external internet users, subject to routing and security rules.

Private IP addresses are commonly used for internal cloud communication because they keep traffic within the cloud network, reduce unnecessary public exposure, and improve network security.

15. A developer manually configures an IP address that is already in use. Describe the symptoms and how you would detect and resolve the conflict.

- An IP address conflict occurs when two devices on the same network are configured with the same IP address.

Symptoms:

- Intermittent network connectivity.
- Duplicate IP address warnings.
- Unstable communication with other devices.
- Packets may reach the wrong device.

Detection:

Use commands such as ip neigh and arping to investigate duplicate IP usage and identify the device responding to the conflicting address.

Resolution

- Identify the devices using the duplicate IP address.
- Assign a unique IP address to one of the devices or configure it to use DHCP.
- Check the DHCP address pool and static IP assignments to prevent future conflicts.
- Verify network connectivity after making the changes.

PART D – DNS Investigation

Run and record: nslookup google.com, dig google.com A, dig google.com MX, and dig +trace
google.com (where available).

16. Compare the purpose and useful output of nslookup and dig.

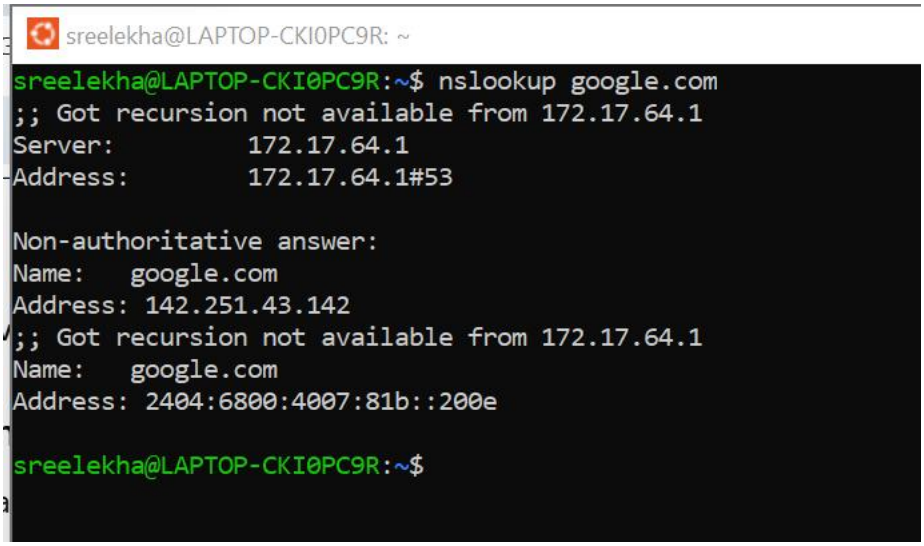
Both nslookup and dig are command-line tools used to investigate DNS information.

nslookup:

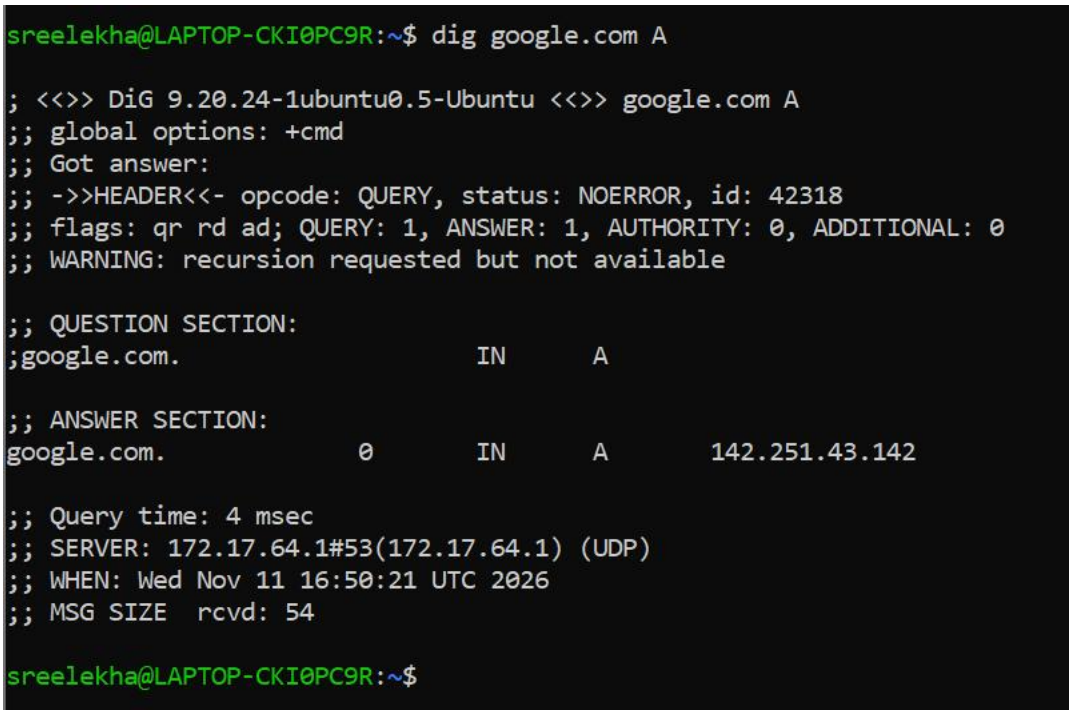
Performs basic DNS queries.

Displays the DNS server used and the resolved IP address.

Is simple and easy to use for basic domain lookups.



dig:



Provides detailed DNS query information.

Displays sections such as QUESTION, ANSWER, AUTHORITY, and ADDITIONAL.

Shows query status, response time, TTL, and DNS server details.

Supports different record types such as A, AAAA, MX, and CNAME.

Conclusion: nslookup is useful for simple DNS lookups, whereas dig is more suitable for detailed DNS troubleshooting and analysis.

17. Explain the difference between an A record, AAAA record, CNAME record, and MX record. Give a realistic use case
for each.

- **A Record:** Maps a domain name to an IPv4 address.
  Example: example.com → 192.0.2.10
  Use case: Directing website visitors to a server using IPv4.
- **AAAA Record:** Maps a domain name to an IPv6 address.
  Example: example.com → 2001:db8::10
  Use case: Making a website accessible through IPv6.
- **CNAME Record:** Maps an alias domain name to another canonical domain name.
  Example: www.example.com → example.com
  Use case: Pointing a www subdomain to the main domain.
- **MX Record:** Specifies the mail servers responsible for receiving email for a domain.
  Example: example.com → mail.example.com
  Use case: Directing incoming emails to the correct mail server.

These records help DNS provide different types of network services.

18. A domain resolves correctly on mobile data but returns an old IP on an office machine. Give at least four possible
causes and a safe verification sequence.

If a domain resolves to the correct IP on mobile data but returns an old IP on an office machine, possible causes include:

- **Local DNS cache:** The office computer may have cached an old DNS response.

- **DNS resolver cache:** The office DNS server may still have the previous record.
- **TTL:** The previous DNS response may still be within its Time To Live period.
- **Hosts file:** A manually configured entry may override DNS resolution.
- **Different DNS resolvers:** The office and mobile networks may use different DNS servers.
- **Split DNS:** The office network may intentionally return an internal IP address.

Safe verification sequence:

- Run nslookup example.com on the office machine.
- Run dig example.com A to inspect the DNS response and TTL.
- Compare the result using a public resolver, for example nslookup example.com 8.8.8.8.
- Check the configured DNS server and hosts file.
- Check the office DNS cache and network policies.
- Clear or refresh the relevant cache only after identifying the cause.

This process helps determine whether the issue is caused by local caching, resolver configuration, TTL, or internal DNS policies.

19. Explain recursive DNS resolution at a high level, including the roles of a resolver, root server, TLD server, and
authoritative name server.

Q19. Explain Recursive DNS Resolution

Recursive DNS resolution is the process of finding the IP address associated with a domain name by querying DNS servers.

The main components are:

1.Recursive Resolver: Receives the client's DNS query, checks its cache, and obtains the answer if necessary.

2.Root DNS Server: Directs the resolver to the appropriate Top-Level Domain (TLD) name server.

3.**TLD DNS Server:** Provides information about the authoritative name server for the requested domain.

4.Authoritative Name Server: Stores the actual DNS records and provides the requested DNS answer.

Resolution process:

When a user requests www.google.com, the recursive resolver checks its cache. If the answer is not cached, it queries the root server, then the appropriate TLD server, and finally the authoritative name server. The answer is returned to the client and may be cached for future queries.

This process allows users to access websites using domain names instead of remembering IP addresses.

20. A DNS record has just been changed. Explain TTL and why different users may temporarily receive different
answers.

TTL stands for Time To Live. It specifies how long a DNS record can remain in a DNS resolver's cache before it should be refreshed.

For example, if a DNS record has a TTL of 3600 seconds, the resolver may cache that record for 60 minutes.

When a DNS record is changed, different users may temporarily receive different IP addresses because:

- Some DNS resolvers may still have the old record cached.
- The old record may not have reached the end of its TTL.
- Different DNS resolvers may refresh their caches at different times.
- Local devices may also have cached DNS responses.

Once the cached records expire and resolvers obtain the updated DNS information, users should receive the new answer. Therefore, DNS changes may take time to appear consistently across different networks.